
```
clc;
clear;
close all;

% Set random seed for repeatable results
rng(1);

% Input Bits
data = randi([0 1], 20, 1);

% Convolutional Encoder
% Rate = 1/2
% Constraint Length = 3
% Generator Polynomials = [7 5] in octal
trellis = poly2trellis(3, [7 5]);

% Perform Convolutional Encoding
encoded = convenc(data, trellis);

% Display Input and Encoded Bits
disp('Input Bits:');
disp(data');

disp('Encoded Bits:');
disp(encoded');

% Display lengths
fprintf('\nNumber of Input Bits: %d\n', length(data));
fprintf('Number of Encoded Bits: %d\n', length(encoded));
fprintf('Code Rate: 1/2\n');

% Plot Input Bit Sequence
figure;

subplot(2,1,1);
stem(data, 'filled', 'LineWidth', 1.2);
title('Input Bit Sequence');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

% Plot Convolutionally Encoded Bit Sequence
subplot(2,1,2);
stem(encoded, 'filled', 'LineWidth', 1.2);
title('Convolutionally Encoded Bit Sequence');
xlabel('Encoded Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

Input Bits:
Columns 1 through 13
```

0	1	0	0	0	0	0	0	0	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---

Columns 14 through 20

1	0	1	0	1	0	0
---	---	---	---	---	---	---

Encoded Bits:

Columns 1 through 13

0	0	1	1	1	0	1	1	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

Columns 14 through 26

0	0	0	0	0	1	1	1	0	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---

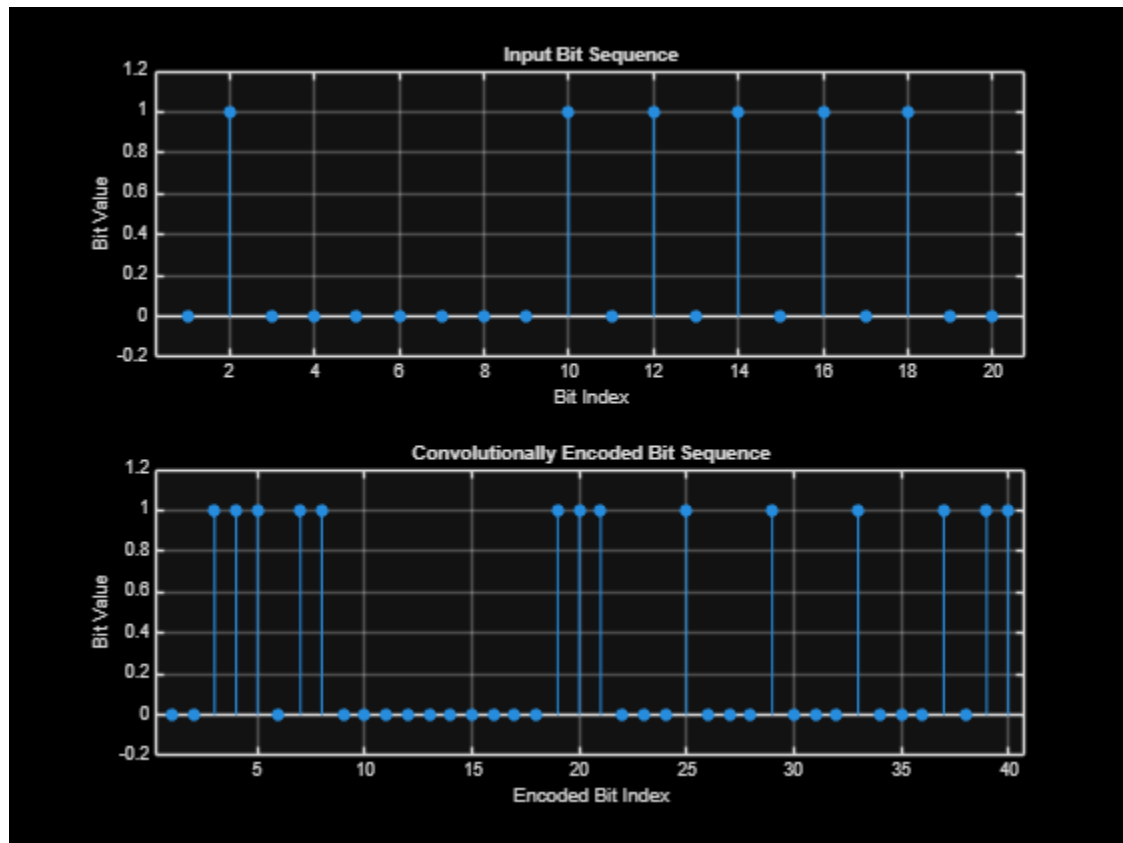
Columns 27 through 39

0	0	1	0	0	0	1	0	0	0	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---

Column 40

1

Number of Input Bits: 20
Number of Encoded Bits: 40
Code Rate: 1/2



Published with MATLAB® R2026b